James A. Barron

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Queen's University, Kingston, Canada
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Education

PhD Candidate, Astrophysics, Queen's University, Kingston, Canada — Jan 2023 – Present (expected Mar 2027)
Thesis: *Spectropolarimetric Studies of Magnetism in Cepheids and Massive Stars*
Supervisor: Gregg A. Wade

MSc, Astrophysics, Queen's University, Kingston, Canada — Dec 2022
Thesis: *A Systematic Study of the TESS Photometry of the Magnetic O-type Stars*

BSc, Physics (Honours), Queen's University, Kingston, Canada

BSc, Specialization in Mathematics, University of Alberta, Edmonton, Canada

Publications

First-Author Refereed Publications

1. **Barron, J. A.**, Wade, G. A., & Folsom, C. P. (2026), The Rotational and Magnetic Properties of Polaris from Long-term Spectropolarimetric Monitoring, ApJ, 1005, 112, (SciX)
2. **Barron, J. A.**, Wade, G. A., Holgado, G., & Simón-Díaz, S. (2025), A Magnetic Field Detection in the Massive O-type Bright Giant 63 Oph, ApJ, 991, 79, (SciX)
3. **Barron, J. A.**, Wade, G. A., Evans, N. R., Folsom, C. P., & Neilson, H. R. (2022), Finding magnetic north: an extraordinary magnetic field detection in Polaris and first results of a magnetic survey of classical Cepheids, MNRAS, 512, 4021–4030, (SciX)

Co-Authored Refereed Publications

1. Biswas, A., Folsom, C. P., **Barron, J. A.**, Wade, G. A., Bellotti, S., & Trigilio, C. (2026), Discovery of a Rapidly Evolving Global Magnetic Field in the M Dwarf YZ Cet and Constraints on the Magnetic Field of Its Planet YZ Cet b, ApJ, 997, 268, (SciX)
2. Biswas, A., Das, B., **Barron, J. A.**, Wade, G. A., & Holgado, G. (2025), A Nonstop Aurora? The Intriguing Radio Emission from the Rapidly Rotating Magnetic Massive Star HR 5907, ApJ, 980, 260, (SciX)
3. Wade, G. A., Oksala, M., Neiner, C., Boucher, É., & **Barron, J. A.** (2025), Magnetic Field Monitoring of Four Massive A–F Supergiants, ApJL, 988, L38, (SciX)
4. Frost, A. J., Sana, H., Mahy, L., Wade, G., **Barron, J.**, Le Bouquin, J.-B., Mérand, A., Schneider, F. R. N., Shenar, T., Barbá, R. H., Bowman, D. M., Fabry, M., Farhang, A., Marchant, P., Morrell, N. I., & Smoker, J. V. (2024), A magnetic massive star has experienced a stellar merger, Science, 384, 214–217, (SciX)

Manuscripts to be Submitted

1. **Barron, J. A.**, Wade, G. A., Folsom, C. P., & Kochukhov, O., First magnetic detection in a W Virginis Type II Cepheid: The shock-affected magnetic signature of κ Pavonis, in preparation for ApJ

Conference Proceedings

1. **Barron, J. A.**, Wade, G. A., Holgado, G., & Simón-Díaz, S. (in press), Discovery of a Magnetic Field in the O-type Bright Giant 63 Oph, IAUS, 402
2. **Barron, J. A.**, Wade, G. A., Folsom, C. P., & Kochukhov, O. (2024), First results of a magnetic survey of classical Cepheids, IAUS, 361, 233–235, (SciX)
3. Frost, A. J., Sana, H., Mahy, L., Wade, G., **Barron, J.**, Le Bouquin, J.-B., Mérand, A., Schneider, F. R. N., Shenar, T., Barbá, R. H., Bowman, D. M., Fabry, M., Farhang, A., Marchant, P., Morrell, N. I., & Smoker, J. V. (2024), An interaction and merger in a massive multiple system create a magnetic field in a massive star, Bulletin de la Société Royale des Sciences de Liège, 93, 85–102, (SciX)
4. **Barron, J.**, Wade, G. A., Bowman, D. M., David-Uraz, A., Munoz, M. S., Pablo, H., & Simón-Díaz, S. (2020), MOBSTER: Identifying Candidate Magnetic O Stars through Rotational Modulation of TESS Photometry,

Proceedings of the Polish Astronomical Society, 11, 226–234, (SciX)

5. David-Uraz, A., Neiner, C., Sikora, J., **Barron, J.**, Bowman, D. M., Cerrahoğlu, P., Cohen, D. H., Erba, C., Khalack, V., Kobzar, O., Kochukhov, O., Pablo, H., Petit, V., Shultz, M. E., ud-Doula, A., & Wade, G. A. (2020), Magnetic OB[A] Stars with TESS: probing their Evolutionary and Rotational properties - The MOBSTER Collaboration, Stars and their Variability Observed from Space, 471–474, (SciX)
6. David-Uraz, A., Neiner, C., Sikora, J., **Barron, J.**, Bowman, D. M., Cerrahoğlu, P., Cohen, D. H., Erba, C., Kobzar, O., Kochukhov, O., Petit, V., Shultz, M. E., ud-Doula, A., Wade, G. A., & MOBSTER Collaboration (2020), MOBSTER: Establishing a Picture of Magnetic Massive Stars as a Population, Proceedings of the Polish Astronomical Society, 11, 219–225, (SciX)

Community Papers

1. **Barron, J. A.**, Wade, G. A., Bagnulo, S., Biswas, A., Bowman, D. M., David-Uraz, A., Erba, C., Eze, C. I., Folsom, C. P., Gutteridge, C., Khalack, V., Krtička, J., Labadie-Bartz, J., Landstreet, J. D., Natan, T., Neiner, C., Oksala, M., Petit, V., Shultz, M. E., Sliusarenko, M., Stanley, P., & Xia, Z. (2024), New capabilities for the Gemini Observatory: The case for high resolution optical spectropolarimetry, Community Brief Recommendation submitted to the Gemini Observatory Strategic Vision process

Theses

1. **Barron, J.** (2023), A Systematic Study of the TESS Photometry of the Magnetic O-type Stars, MSc Thesis, Queen's University, (SciX)

Invited Talks

Royal Military College of Canada, Kingston, Canada — Feb 2027 (scheduled)
The magnetic properties of evolved massive and pulsating stars

Instituto de Astrofísica de Canarias, Tenerife, Spain — Oct 2022
Magnetic properties of classical pulsators

Contributed Talks

MIAPbP: Bridging Gaps in High-Mass Star Research, Garching, Germany — Aug 2026
Magnetism of massive stars: Recent results and open questions (with G. A. Wade)

CASCA 2026: MIL-ion Futures of Astrophysics, Montréal, Canada — Jun 2026
The rotational and magnetic properties of Polaris: Challenges to standard stellar models

14th CFHT Users' Meeting, Lac-à-l'Eau-Claire, Canada — May 2025
Stellar magnetism in the instability strip: New magnetic detections in classical and Type II Cepheids

MiMeS Workshop, Kingston, Canada — May 2025
Follow-up of MiMeS magnetic O star candidates

MIAPbP Stellar Magnetic Fields Workshop, Garching, Germany — Oct 2023
Stellar magnetism through the instability strip

TASC6/KASC13 Workshop, Leuven, Belgium — Jul 2022
A systematic study of the TESS photometry of the magnetic O-type stars

MOBSTER-1 Virtual Conference, Virtual — Jul 2020
MOBSTER: Identifying Candidate Magnetic O Stars through Rotational Modulation of TESS Photometry

Poster Presentations

First-Author Posters

IAU Symposium 402: Massive Stars Across Redshifts, Ensenada, Mexico — Sep 2025
Discovery of a Magnetic Field in the O-type Bright Giant 63 Oph

Magnetic Fields from Clouds to Stars (Bfields-2024), Tokyo, Japan — Mar 2024
The magnetic fields of pulsating Cepheid variables (poster and flash talk)

RR Lyrae & Cepheid Stars Conference, La Palma, Spain — Sep 2022
First results of a magnetic survey of classical Cepheids

IAU Symposium 361: Massive Stars Near and Far, Ballyconnell, Ireland — May 2022
First results of a magnetic survey of classical Cepheids

CASCA 2022: Canadian Astronomy in the Roaring 2020's, Virtual — May 2022
Finding magnetic north: an extraordinary magnetic field detection in Polaris and first results of a magnetic survey of classical Cepheids

13th CFHT Users' Meeting, Virtual — May 2022
Finding magnetic north: an extraordinary magnetic field detection in Polaris and first results of a magnetic survey of classical Cepheids

OBA Stars: Variability and Magnetic Fields, Virtual — Apr 2021
TESS photometry of the Of?p stars HD 148937 and LMC 164-2 (flash talk)

MOBSTER-1 Virtual Conference, Virtual — Jul 2020
Is the magnetic activity of Classical Cepheids modulated by pulsation? The case of η Aql

Awards and Funding

Natural Sciences and Engineering Research Council of Canada (NSERC) PGS-D — 2023 – 2026
Queen's University

G. Neilson Whyte Graduate Fellowship, Queen's University — 2025

Conference Travel Award, Queen's University (3 awards) — 2022 – 2025

Harold M. Cave Graduate Travel Scholarship, Queen's University (3 awards) — 2023 – 2025

R. S. McLaughlin Fellowship, Queen's University — 2022 – 2023

IAU Travel Grant, International Astronomical Union (IAU Symposium 361) — 2022

Carl Reinhardt Fellowship, Queen's University — 2021 – 2022

Graduate Entrance Tuition Award, Queen's University — 2019 – 2020

NSERC Undergraduate Student Research Award, Royal Military College of Canada — 2019

Dean's Honour List, Queen's University — 2018 – 2019

Harold M. Cave Undergraduate Travel Scholarship, Queen's University — 2018

Research Visits and Workshops

Instituto de Astrofísica de Canarias, Tenerife, Spain — Nov 2026 (scheduled)
Research visit (1 week)

Munich Institute for Astro-, Particle and BioPhysics, Garching, Germany — Aug 2026
Selected participant, *Bridging Gaps in High-Mass Star Research* (2 weeks)

Munich Institute for Astro-, Particle and BioPhysics, Garching, Germany — Oct 2023
Selected participant, *Stellar Magnetic Fields from Protostars to Supernovae* (2 weeks)

Instituto de Astrofísica de Canarias, Tenerife, Spain — Oct 2022
Research visit (2 weeks)

Research Mentorship

Designed and co-supervised research projects for 6 undergraduate students. Outcomes include an undergraduate thesis, an open-source analysis pipeline, final reports and presentations.

Undergraduate Research (3 students), Royal Military College of Canada — May 2026 – Aug 2026
Development of an open-source software analysis pipeline for Gaia DR4

Undergraduate Research Thesis, Queen's University — Sep 2025 – Apr 2026
Inferring the atmospheric velocity gradients of classical Cepheids

Undergraduate Research (NSERC USRA), Royal Military College of Canada — May 2024 – Aug 2024
A first BRITE, SMEI and TESS analysis of the suspected merger product θ-Scorpii
Talk by G. A. Wade, 2024 BRITE Side of Stars Conference, Vienna, Austria, (SciX)

Undergraduate Research (NSERC USRA), Royal Military College of Canada — May 2023 – Aug 2023
Searching for slowly rotating magnetic massive stars with Gaia

Teaching Experience

Led laboratory and tutorial sections and marked assignments and exams for first-year through fourth-year undergraduate physics and astronomy courses (18 appointments, 9 courses).

Teaching Assistant, Royal Military College of Canada — Sep 2026 – Present
General Physics (PHE 104)

Teaching Assistant, Queen's University — Sep 2019 – Present
Stellar Structure and Evolution (PHYS 435); Physical Processes in Astrophysics (PHYS 315); Experimentation and Design (APSC 102); The Solar System (ASTR 101); Stars, Galaxies, and the Universe (ASTR 102); Introduction to Physics I (PHYS 115); Introduction to Physics II (PHYS 116); Introductory Physics Laboratory (PHYS 117)

Professional Service and Leadership

Scientific Organizing Committee, CASCA Emerging Researchers Showcase — Aug 2026 – Oct 2026
Graduate Student Committee Representative, Canadian Astronomical Society — Sep 2022 – Aug 2026
Graduate Oral Presentation and Poster Judge, CASCA 2026 — Jun 2026
Presenter, Queen's / RMC Astronomy Research Group Journals Club — Sep 2019 – Mar 2026
Adjudicator, Inquiry@Queen's Undergraduate Research Conference — Mar 2026
Graduate Poster Judge, CASCA 2025 — Jun 2025
Local Organizing Committee and Session Chair, MiMeS Workshop, Kingston — May 2025
Graduate Steering Representative, Queen's Graduate Physics Society — Sep 2022 – Aug 2023
Adjudicator, Inquiry@Queen's Undergraduate Research Conference — Mar 2022
Local Organizing Committee and Session Chair, MOBSTER-2 Virtual Workshop — Nov 2021
Conference Volunteer, The Physics of Galaxy Scaling Relations, Queen's University — Jul 2018

Public Outreach

Observatory Volunteer, Queen's Observatory — Sep 2023 – Present
Volunteer, Queen's Astronomy on Tap — Sep 2023 – Present
Public Talk, Royal Astronomical Society of Canada, Kingston Centre — Oct 2026
The Lives of Magnetic Massive Stars

Public Talk, Queen's Observatory — Sep 2026
Cosmic Engines: The Lives and Deaths of Massive Stars

Astronomy Exhibit Planning and Volunteer, Science Rendezvous Kingston — Oct 2025 – May 2026
Category Judge, Frontenac, Lennox & Addington Science Fair — 2021, 2022, 2025, 2026
Volunteer, Science Literacy Week, McDonald Institute & Queen's Observatory — Oct 2025
Outreach Volunteer, Science Rendezvous Kingston — 2024, 2025
Eclipse Ambassador, Queen's University — Sep 2023 – Apr 2024
Research Feature, Journal of the Royal Astronomical Society of Canada — Jun 2022
CFHT Chronicles: Diving into a Magnetic Polaris by Mary Beth Laychak, (SciX)

Interviewed, CFRC-FM Radio, Kingston, Canada — Apr 2022
Science Fair Judge, Canada-Wide Science Fair — May 2021
Let's Talk Astrophysics Volunteer, Let's Talk Science, Queen's University — Feb 2021
Outreach Presentation, Greenview Elementary School, Edmonton, Canada — Dec 2020

Professional Development

Canary Islands Winter School, Massive Stars as Cosmic Tools — Nov 2026 (scheduled)
Instituto de Astrofísica de Canarias, Tenerife, Spain

Research Project Management Training — Jan 2026
Canadian Association for Graduate Studies

Science Leadership and Management — Dec 2022
Queen's University (semester course)

Introduction to Astronomical Instrumentation — Jul 2021
Dunlap Institute Summer School, University of Toronto

Summer School on High Performance and Technical Computing — Aug 2019
Centre for Advanced Computing, Queen's University

CRAQ Summer School on Stellar Astrophysics — Jun 2019
Centre for Research in Astrophysics of Québec, McGill University

Observing Programs

PI on 20 accepted proposals totalling over 150 h at CFHT, the ESO 3.6-m telescope and Gemini Observatory. Co-I on 17 further proposals, including a funded NASA ADAP grant and PLATO, GMRT and NICER observing programs. Observing experience as a visiting astronomer at La Silla Observatory (ESO 3.6-m/HARPSpol). Select programs listed.

Principal Investigator

CFHT/ESPaDOnS, 27 h over 7 semesters (incl. DDT) — 2021 – 2026
Finding Magnetic North: A first magnetic map of the classical Cepheid Polaris

CFHT/ESPaDOnS, 14 h — 2024
Concluding the first systematic magnetic study of classical Cepheids: probing the low and high mass regions of the instability strip

ESO 3.6-m/HARPSpol, 38 h over 2 semesters, visitor mode — 2021 – 2023
A first systematic study of the magnetic fields of classical Cepheid stars

CFHT/ESPaDOnS, 54 h over 4 semesters — 2020 – 2022
A first systematic study of the magnetic fields of classical Cepheid stars

CFHT/ESPaDOnS, 7 h — 2022
Follow-up observations of two magnetic O star candidates identified in the MiMeS survey

Co-Investigator

PLATO, 2-yr Guest Observer programme, AO-1 (PI: C. Neiner) — 2027 – 2029
What is the internal magnetic field of pulsating hot (β Cep, SPB, δ Scu) stars?

PLATO, 2-yr Guest Observer programme, AO-1 (PI: K. Thomson-Paressant) — 2027 – 2029
The relationship between magnetism and pulsations in classical pulsators

CFHT/SPIRou, 3 h (PI: A. Biswas) — 2024
Constraining the magnetic field of YZ Cet b through star-planet Interactions

GMRT, 14 h (PI: A. Biswas) — 2024
Follow-up uGMRT observation of the most rapidly-rotating, non-degenerate, magnetic massive star: HD 142184

Technical Skills

Spectropolarimetry: ESPaDOnS/NARVAL reduction and analysis, Least-Squares Deconvolution (ILSD, SpecpolFlow), Zeeman Doppler Imaging (ZDIpy), magnetic field diagnostics

Spectroscopy: Spectrum synthesis and line-broadening analysis (ZEEMAN, IACOB-BROAD), radial velocities, binary orbit modelling, line bisectors, telluric correction (TelFit)

Photometry and time-series analysis: Custom TESS light-curve extraction from target pixel files and full-frame images (Lightkurve, TESScut), K2 halo photometry (halophot), Gaia epoch photometry, Hipparcos and ASAS-SN light curves, custom prewhitening pipeline (Lomb–Scargle, lmfit), Period04

Statistics and programming: Python (NumPy, SciPy, Astropy, Astroquery, lmfit, emcee, corner, specutils), MCMC and Monte Carlo methods, IDL (basic), Fortran (basic)

Observing: Visiting astronomer at the ESO 3.6-m telescope (HARPSpol), La Silla Observatory

Collaborations and Professional Memberships

Amakihi, CFHT optical and infrared spectropolarimetric community survey — 2025 – Present

MUDEHaR, Hα ground-based photometric survey — 2023 – Present

MOBSTER Collaboration — 2019 – Present
Magnetic OB[A] Stars with TESS: probing their Evolutionary and Rotational properties

Canadian Astronomical Society (CASCA) — 2022 – Present